Architectural Requirements		
ID	Requirement	Architecture
(FR_NOWEAPON)	No injury or violence	No blood, no graphical violence - instead it will be food that deters the geese, ‘distract’ elements rather than attack. Agitation is its own class and a full meter triggers the tantrum delay.
(UR_NOHARM)	No health bar to indicate harm or healing	There will be an ‘agitation’ bar, once full it will lead to an event ‘tantrum’ that delays your progress in clearing the geese
(UR_COUNTER) (UR_EVENTS) (UR_MAP) (UR_ITEM)	Entertainment and Fulfilment	To ‘pad out’ the game, we will add other animals to feature periodically. They may not be interactable, or perhaps they will interact with items such as stealing bread that will delay your progress. This may depend on difficulty settings.
(UR_COMPUTE)	Full screen and windowed launcher or window configuration.	Says where the fullscreen and windowed modes are set.
(UR_CONTROL)	Avatar that the player controls	The avatar is represented by the Player class, which holds its position and movement speed.
(UR_TIMER)	Active countdown timer	The timer is a single class that owns the countdown. Events have no influence over the timer, which keeps the game in real time.
(UR_START)	Evidence of a start/ title screen	A separately designed title screen. This is the screen shown when the game launches.
(UR_P_MOVE)	Visual of the character moving within the environment	The avatar moves in synch with the keyboard movements (arrows, WASD).
(UR_COUNTER)	Visual of the item counter	A graph with slots that fill with images of the interactables as you encounter them
(UR_EVENTS)	Hidden events that create visuals and interactables	

(UR_INV) (UR_ITEM)	Interactable inventory items Visibly unique items to distract the geese with	The player should have visually unique items that they can interact with. The inventory is created with two items, and using an item removes it and triggers the goose's reaction. Item types as subclasses or data make new items easy to add.
(UR_MAP)	The map that the game takes place within	A clear detail of where the map ends and where the player can explore. Design features may be added such as nature or foliage to indicate the edge of the map and clearly tell the player where to go.
(UR_START)	Start screen	Each screen has its own class, so menus and gameplay are independent.
(UR_VIEW)	Top Down	A fixed, top-down camera that renders the map and all entities from above. This allows for a consistent view of all interactables while keeping the game design simple.
(FR_START_PAUSE) FR_PAUSEKEY)	Every round begins with the paused state until the player resumes the timer	An explicit state means the timer and entities only update while running, so each round starts paused, and the ESC toggles the state.
(FR_GEESE)	Visual of geese chasing the player	Chase behaviour lies within the goose itself, ideal for the range of behaviour: idle, chasing, distracted. Visual effects to indicate the change in state.
(FR_EVENT_EFFECT) (FR_EVENT_ONCE) (FR_EVENTTIME)	Visual of an icon to represent a good or bad impact on the player after an interaction or event No visualisation is triggered after the same event has already happened Counter remains consistent regardless of events	Each event stores a triggered flag so it fires once, and it carries its effect type (good or bad) for the icon.
(FR_INVINT)	The inventory slots will have two items at the beginning of the game	The inventory is visible on the side of the screen so the player can see which items are picked up and when they are used.
(FR_ITEMUSE)	The items are removed once used	Items disappear after use.

(FR_ITEMEFFECT)	A react or emote when an item is used against a goose	When an item is used, it calls a reaction on the targeted goose, which switches to a ‘distracted’ state and displays an emote.
(FR_BOUNDARY)	The player and geese must only be able to move within a clearly marked parameter, and not outside the map.	Stores the map boundary as a collision layer. Both the player and goose check a move against the map before making it, so the boundary rule is defined once and applies to every moving entity.
(FR_WIN) (FR_RESULTSCREEN) (FR_SCORE) (FR_TIMEDISPLAY)	A visual of a win event once the game is complete. This will detail the remaining time as well as the score of the player’s round.	When the win condition is met, the game passes the remaining time through the timer and score then displays the result.
(FR_TIMEOUT)	A timeout screen for when the player loses	When the timer reaches zero, the screen switches to the result screen in its timeout mode. Win and lose share one screen class with different content, which avoids duplicated layout code.
(FR_VISUALS)	Clear visuals of avatar, geese, interactables.	Each player, geese, and item has its own distinct sprite, loaded once at start-up. The geese spawn in gradually. The visuals from hidden events only appear once triggered.
(QR_FPS)	System running at 60fps for a smooth interface and no lagging on low system requirements	The game loop updates and renders once per frame, capped at 60fps and using delta time. This is to prevent the game lagging if the frame rate dips on lower spec computers.

The table shows how each architectural design maps on to the system and design requirements. The architecture is organised so that each group of related requirements is the responsibility of one construct or group of ID’s.

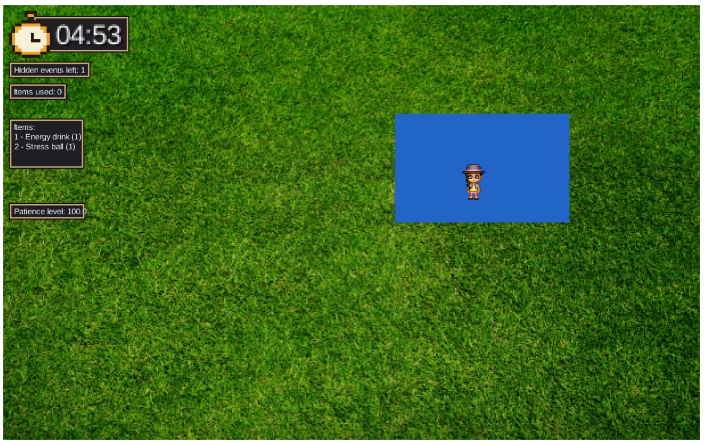
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Title Screen



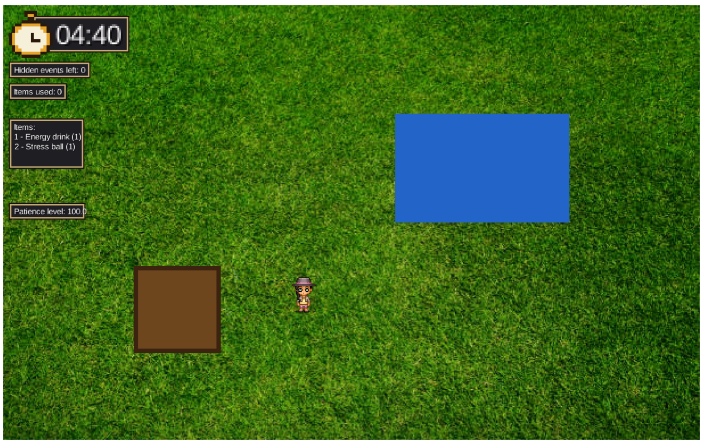
Main menu screen. Start Game, Restart Game, Exit.

Lake Event

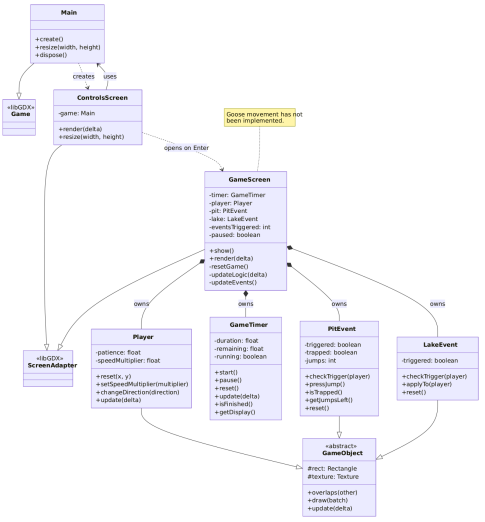


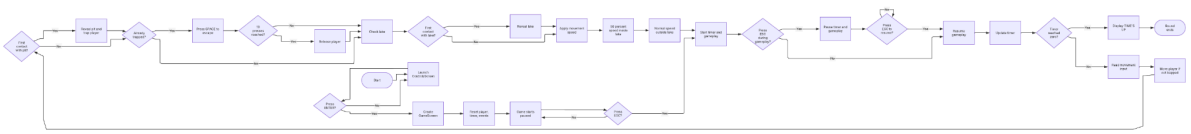
Lake is triggered when the player explores an area of the map.

No Events Repeated after being triggered once



This flow diagram elaborates on how the game behaves when items and events are triggered. **Start/End** - This shows where the process starts and ends. **Action (Rectangle)** - Represents something happening, such as moving the player or starting the timer. This is like an attribute. **Diamond** - This represents a decision or question.





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Yes or No branches.

- If yes - then details the events that follow one particular event
- If no - then any other possible event/ interaction

Arrow - Shows which direction the process follows

Pit Event



+5 PitEvent. When pit is triggered - use Space key 30 times to escape.

Pause



The game launches in a paused state, enabling the player to manually start the timer.

Time Elapse



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